

Optimizing workflows through an AI-enhanced task management digital solution for effective and sustainable task management

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Abstract

Efficient task management is important to achieve personal and organizational goals. Especially in dynamic environments like in Higher Education (HE), streamlined workflows can help optimize resources and improve collaboration.

This vision abstract explores the design of an AI-enhanced, context-aware task management application aimed at increasing productivity and user experience (UX). This may lead to improvements in sustainability, including innovative sustainable practices in HE (SDG 9). The vision of this research is to explore how the combination of Machine Learning (ML) and context-awareness can enhance UX, improve user productivity, and lead to the adoption of more sustainable practices.

The app will utilize ML to recommend tasks based on user behaviour patterns, while context-awareness will enable in-context reminders to help users stay productive with timely, relevant notifications. The integration of digital calendars may enhance efficiency, reduce disruptions, and promote more focused work sessions. Moreover, the use of multi-view layouts may offer flexibility in work styles, enhancing time and resource allocation. Features such as a task dependency mechanism and analytics could identify workflow bottlenecks, leading to efficiency improvements by promoting user reflection. The app will be evaluated through a quasi-experimental approach focusing on usability and effectiveness in terms of user productivity.

Particularly in HE, this system could serve as a tool for institutions to tackle various challenges relevant to Sustainable Development, such as optimizing resource allocation and distribution, improving productivity by using the system to minimize redundancies and inefficiencies, eliminating biases, and providing information to a diverse set of users (SDG 10).

Keywords: Context-awareness, Machine Learning, Task management, Sustainability, User Experience, Productivity analysis, Resource optimization, Digital transformation